

FMMS

FixFlow Maintenance Management System

Capstone Project — Sprint 1 Deliverable

FULL-STACK WEB APPLICATION

SPRINT 1



Problem Statement & Project Overview

The Problems We're Solving

- Organizations struggle with **high volumes of maintenance requests** managed entirely by hand
- No centralized tracking system for ticket status and history
- Lack of SLA monitoring causes **delayed resolutions** and missed deadlines
- No role-based access control leads to **critical security gaps**
- Inefficient financial tracking for maintenance costs and invoicing

The FMMS Solution

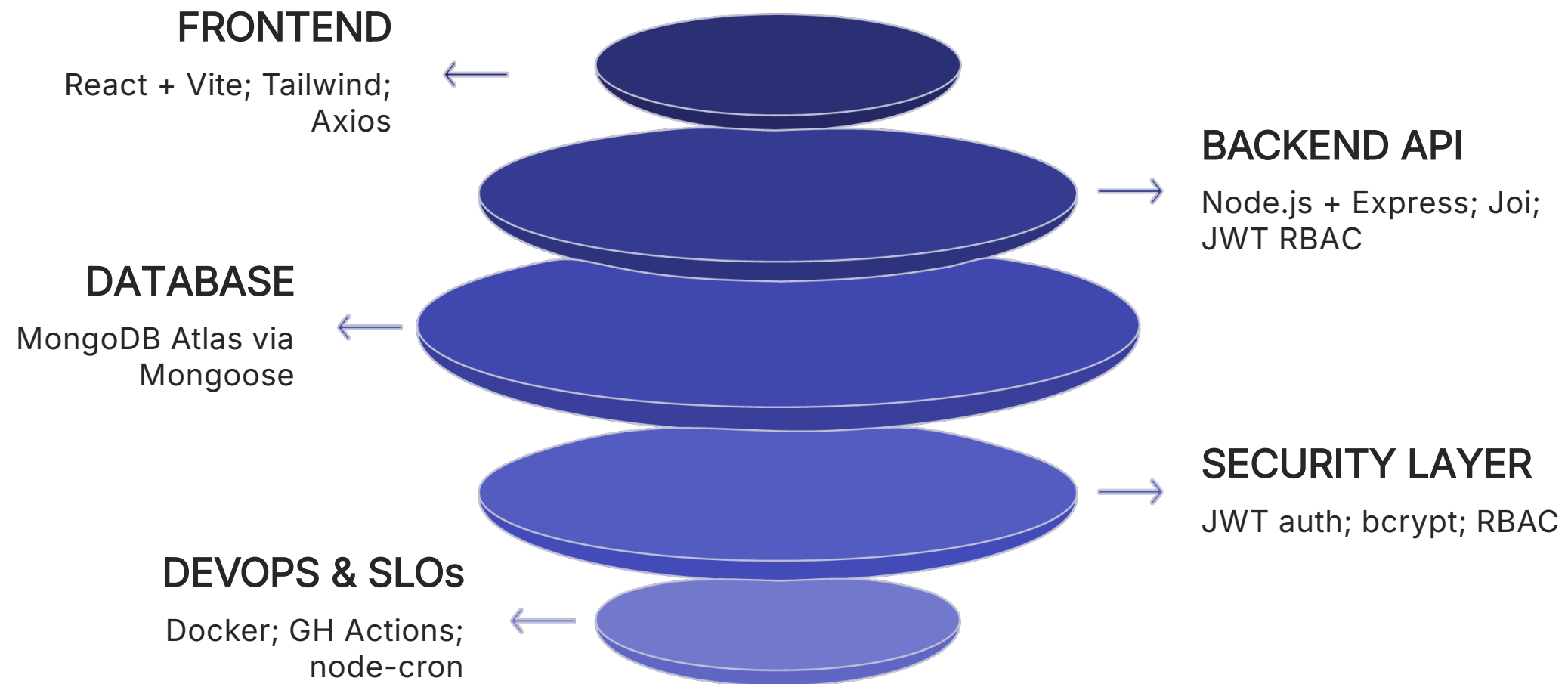
FOMMS is a full-stack web application purpose-built to eliminate these inefficiencies:

- **14-status state-machine-driven** ticket lifecycle
- RBAC across **3 roles**: Admin, Technician, User
- SLA monitoring with **automated escalation alerts**



System Architecture

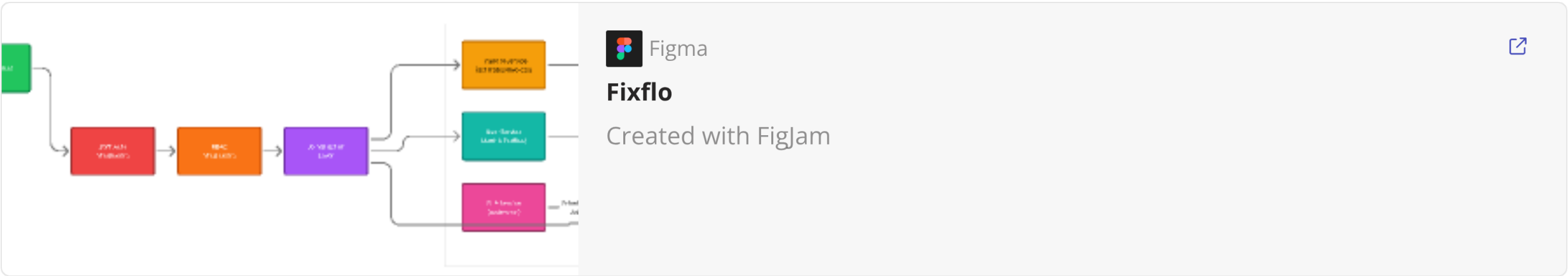
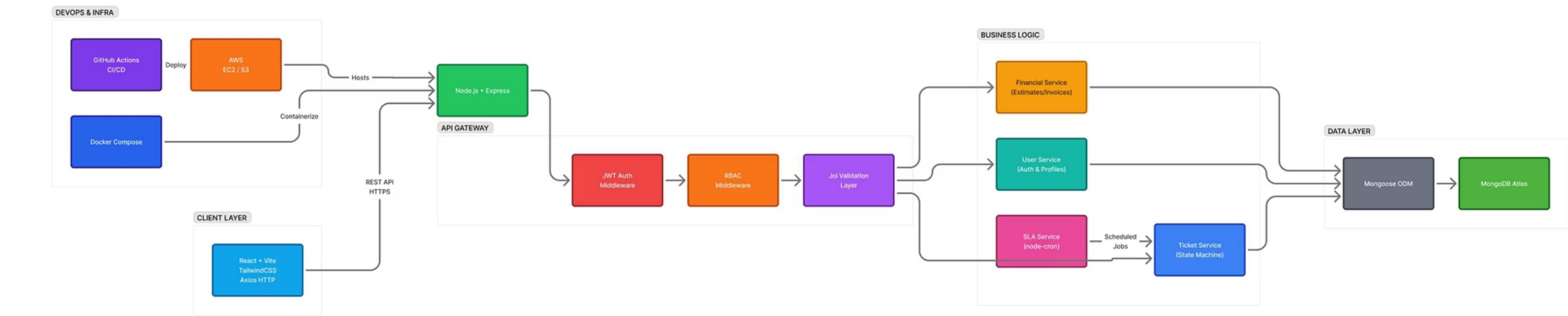
FMMS is built on a modern three-tier architecture with dedicated supporting service layers for security, scheduling, and DevOps automation.

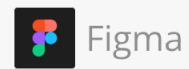
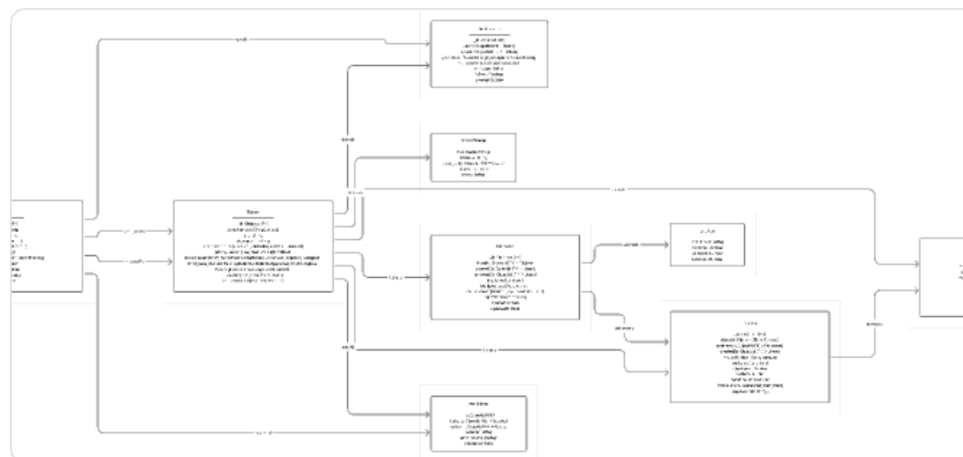
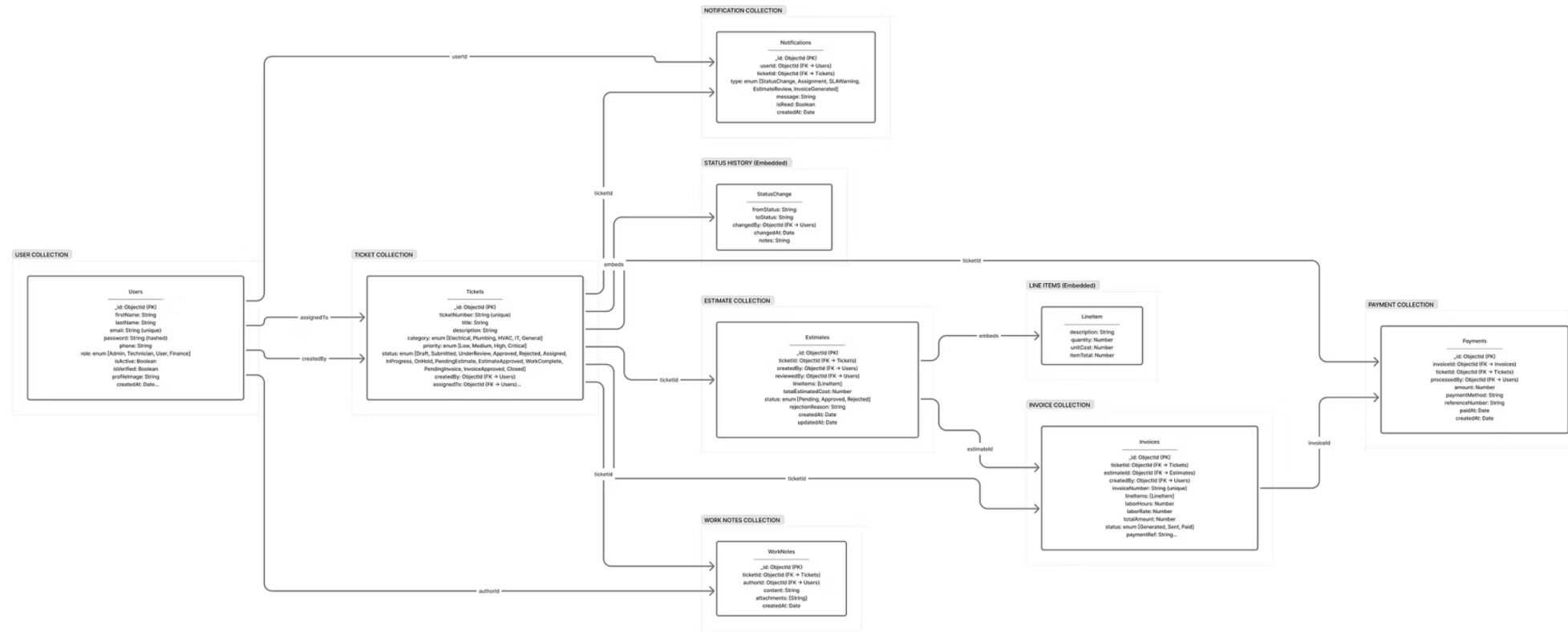


The frontend communicates exclusively through a secured REST API, with all requests passing through JWT authentication and RBAC middleware before reaching the database layer. Supporting services run independently to handle scheduled SLA checks and deployment automation.

Diagrams

Visual representation of the system architecture and interactive components.





fixflo

Created with FigJam



Ticket Lifecycle State Machine

FMMS implements a robust **14-status state machine** that governs every ticket from creation to closure. Each transition is validated server-side, ensuring no illegal status changes occur and maintaining a complete audit trail.



6 Primary States

Draft through In Progress — the core workflow path

3 Branch States

Rejected, On Hold, Pending Estimate — conditional paths

5 Completion States

Estimate Approved through Closed — financial closure



Role-Based Access Control (RBAC)

FMMS enforces strict permission boundaries across four distinct roles. Every API endpoint validates the requesting user's role before executing any operation, ensuring zero unauthorized access.



Admin

- Full system access & configuration
- Manage all users & role assignments
- View all tickets and system reports
- Assign technicians to tickets
- Approve or reject any workflow step



Technician

- View only assigned tickets
- Update ticket status with notes
- Add work notes and attachments
- Submit cost estimates for review
- Mark work complete when done



User

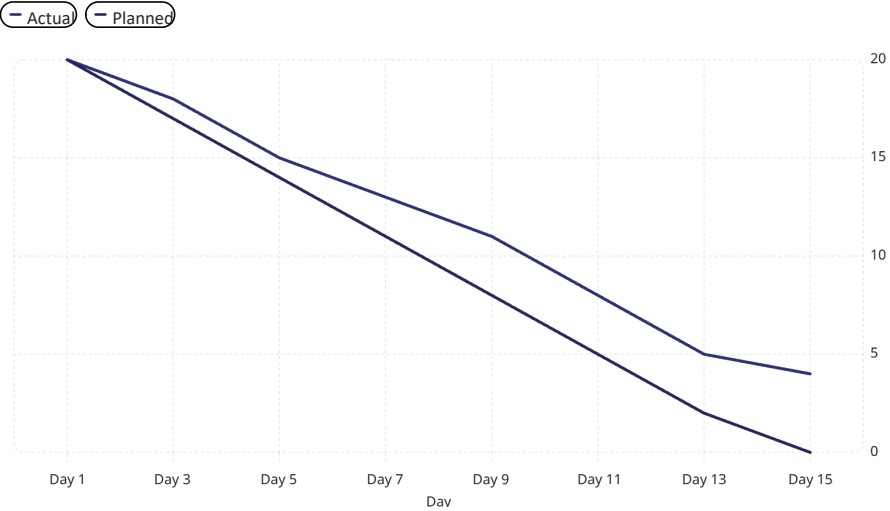
- Create and submit maintenance tickets
- Track status of own tickets
- View full ticket history
- Update personal profile
- Download ticket reports

Sprint Backlog & Team Velocity

Sprint 2 Backlog

ID	Task	Status
SB-1	Create mock-up designs	✅ Done
SB-2	Research React/Vite + MongoDB	✅ Done
SB-3	Complete Sprint 2 presentation	✅ Done
SB-4	Complete user stories	✅ Done
SB-5	Complete acceptance criteria	✅ Done
SB-6	Implement JWT + RBAC auth	✅ Done
SB-7	Build ticket state machine (14 states)	✅ Done
SB-8	Build Admin dashboard	🔄 In Progress
SB-9	Develop financial module	🔄 In Progress
SB-10	Implement SLA monitoring	📋 To-Do
SB-11	Deploy to AWS with Docker	📋 To-Do

Sprint Burndown — 15 Days



10

Sprint 1 Tasks

15

Sprint 2 Tasks

+50% velocity improvement

MVP Features & Technology Stack

Core MVP Features

- **Authentication & Security**
JWT login, registration, password reset, full RBAC enforcement on all endpoints
- **Ticket Management**
14-status state machine, creation & tracking, technician assignment, SLA monitoring
- **Role Dashboards**
Dedicated views for Admin, Technician, User, and Finance roles

Technology Stack

Frontend

React · Vite · TailwindCSS · Axios

Backend

Node.js · Express · Mongoose · Joi

Data & Auth

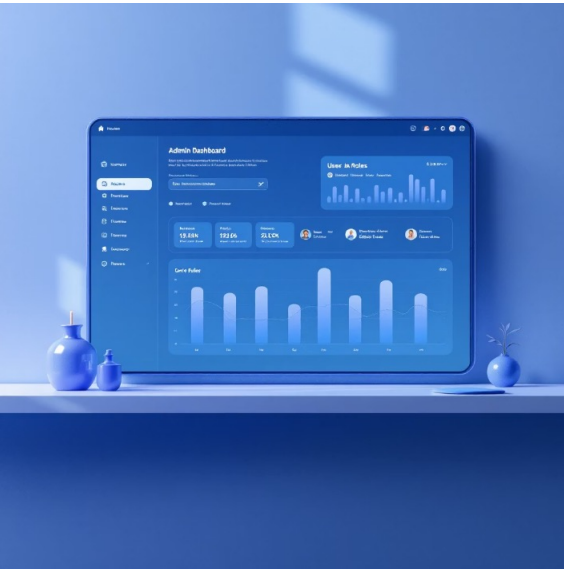
MongoDB Atlas · JWT · bcrypt · node-cron

DevOps & Cloud

Docker Compose · GitHub Actions · AWS



User Stories & Acceptance Criteria



Admin Stories

Manage users/roles, view all tickets, assign technicians

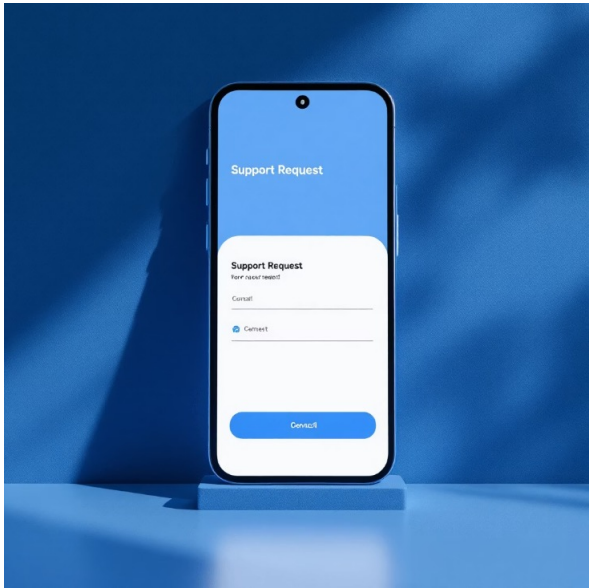
Criteria: Full CRUD works, filters/sorting applied, notifications sent on assignment



Technician Stories

View assigned tickets, update status, submit cost estimates

Criteria: Only sees own tickets, valid transitions enforced, estimates routed for approval



User Stories

Register/login, create tickets, track real-time status

Criteria: Form validation passes, email verification required, status updates visible

Key Test Cases

1

Registration

Validates all fields, email format, and password strength requirements

2

Login

Credential check, error handling, role-based redirect

3

Ticket Creation

Required fields enforced, defaults to Draft status, appears in list view

4

State Transitions

Only valid moves permitted server-side, notifications dispatched on each transition

Sprint Retrospective & UI Mock-ups

What Went Well

- Solid core architecture state machine and RBAC are well-tested and stable
- Team collaboration improved significantly across sprint
- Professional documentation produced and maintained

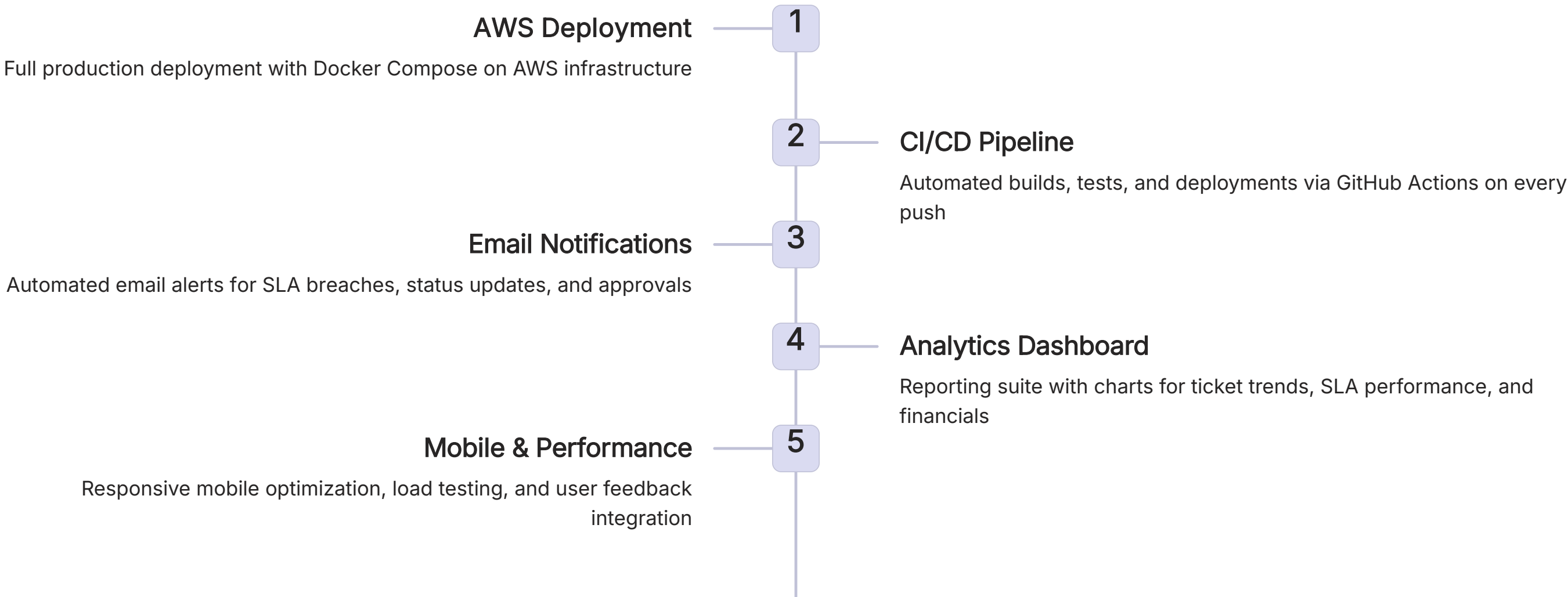
Needs Improvement

- Better coordination needed on frontend-backend integration timing
- More thorough testing of state transition edge cases required

What's Next

- Implement SLA monitoring with node-cron
- Deploy to AWS with Docker Compose
- Set up full CI/CD pipeline via GitHub Actions

Future Roadmap & Next Steps



Acceptance Criteria: Admin & Technician

1

ADMIN ROLE

ID 1: User Management

- Can add, delete, and update user and technician profiles.
- Can assign and modify user roles.

ID 2: Comprehensive Ticket View

- Can view all tickets with advanced filters (status, priority, date).
- SLA status indicators clearly display health (on track, at risk, breached).

ID 3: Technician Assignment

- Can assign technicians based on specialization and availability.
- Assigned technician receives instant notification upon assignment.

2

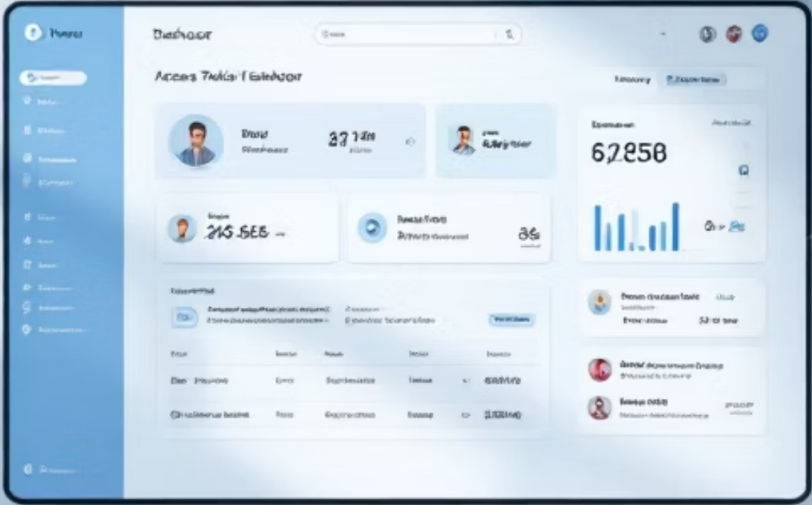
TECHNICIAN ROLE

ID 4: Assigned Ticket View

- Sees only tickets that are explicitly assigned to them.
- Each assigned ticket displays an SLA deadline with a countdown timer.

ID 5: Status Updates & Notes

- Can transition tickets through valid state changes only.
- Ability to add timestamped work notes, including attachments.



Acceptance Criteria: User

1

 USER ROLE

ID 7: Registration

- Registration form validates all required fields.
- Email verification link required for access.

ID 8: Create Tickets

- Can submit ticket with title and priority.
- Can attach files up to 10MB.



Test Cases: Authentication & Registration

ID	Test Scenario	Expected Result
TC-1	Sign Up: Click signup button	Form opens successfully.
TC-2	Sign Up: Submit empty form	Validation errors on required fields.
TC-3	Sign Up: Invalid email format	Email validation error displayed.
TC-4	Sign Up: Password < 8 chars	Strength error shown.
TC-5	Sign Up: Mismatched passwords	Error: Passwords do not match.
TC-6	Sign Up: Valid data	Account created, verification email sent.
TC-7	Sign Up: Existing email	Duplicate email error displayed.
TC-8	Sign In: Open login page	Form renders with email/password fields.
TC-9	Sign In: Incorrect email	"Invalid credentials" error.
TC-10	Sign In: Incorrect password	Error with forgot password link.
TC-11	Sign In: Correct credentials	Redirected to dashboard.
TC-12	Sign In: Unverified email	"Please verify your email" message.
TC-13	Sign In: Forgot password	Password reset email sent.

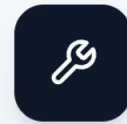
Test Cases: Ticket Creation

ID	Test Scenario	Expected Result
TC-14	Click "New Ticket" button	Ticket creation form opens with all fields.
TC-15	Submit with empty required fields	Validation errors on title, description, category.
TC-16	Fill all fields and set priority to "High"	Ticket created with Draft status.
TC-17	Confirm ticket submission (Draft → Submitted)	Status moves from Draft to Submitted, confirmation shown with ticket ID.
TC-18	Attach file over 10MB	File size limit error displayed.
TC-19	Check ticket list after creation	New ticket appears in user's ticket list.

Test Cases: State Machine Transitions

ID	Test Scenario	Expected Result
TC-20	Admin changes ticket status to "Assigned"	Technician receives notification, ticket status is "Assigned".
TC-21	Admin assigns ticket to specific technician	Assigned technician sees ticket in their queue.
TC-22	Technician changes ticket status to "In Progress"	Ticket status updates to "In Progress", admin and user are notified.
TC-23	Technician adds work notes and attachments	Notes and attachments are saved and visible on the ticket.
TC-24	Technician attempts invalid transition (e.g., "Assigned" to "Resolved")	Error message: "Invalid state transition".
TC-25	Finance approves estimate for a ticket	Ticket status changes to "Approved Estimate".
TC-26	Finance rejects estimate for a ticket	Ticket status changes to "Rejected Estimate", user is notified.
TC-27	Finance generates invoice from approved estimate	Invoice is generated, ticket status updates to "Invoiced".
TC-28	Any role attempts an unauthorized state change	"Permission Denied" error message displayed.
TC-29	Verify all state changes are logged	An auditable trail of all state transitions is available.

Demo UI Images



SOMMS

Smart Organization Maintenance Management

Email

admin@somms.com

Password

.....

Sign In

Default: admin@somms.com / Admin@123456

Welcome back, System

Here's what's happening with your maintenance operations

TOTAL TICKETS

2

SLA COMPLIANCE

100%

AVG RESOLUTION

0h

REVENUE

₹0

Tickets by Status

BILLED1

REVIEWED1

Tickets by Priority

0LOW

1MEDIUM

0HIGH

1EMERGENCY

Recent Tickets

View All >

TCK-2026-0002MEDIUM

There is technical issue going on my floor

REVIEWED>

TCK-2026-0001EMERGENCY

There is electrical problem

BILLED>

Users

Manage system users and roles

NAME	EMAIL	PHONE	ROLE	STATUS
A Amit Finance	amit@somms.com	+91-9876543212	FINANCE	Active
A Anita Verma	anita@somms.com	+91-9876543214	USER	Active
P Priya Sharma	priya@somms.com	+91-9876543211	USER	Active
S Suresh Mehta	suresh@somms.com	+91-9876543213	TECHNICIAN	Active
R Rajesh Kumar	rajesh@somms.com	+91-9876543210	TECHNICIAN	Active
S System Admin	admin@somms.com	+1234567890	ADMIN	Active

Invoices

Billing and payment management

INVOICE #	TICKET	AMOUNT	TAX	TOTAL	STATUS	DUE DATE	
INV-202603-0001	TCK-2026-0001	₹70	₹12.6	₹82.6	GENERATED	4/5/2026	Record Payment

Create New Ticket



Title

Brief description of the issue

Description

Detailed description...

Category

ELECTRICAL



Priority

MEDIUM



Location

Select location...



Create Ticket

Cancel

SOMMS

Dashboard

Tickets

Users

Locations

Invoices

Analytics

Notifications

3

System Admin

ADMIN

Logout

Back to Tickets

TCK-2026-0001

EMERGENCY

BILLED

SLA BREACHED

There is electrical problem

Description

Fix the base problem

Estimates

APPROVED

₹70

Visit: ₹70

Labor: ₹0

Material: ₹0

Other: ₹0

Status History

CREATED

from NONE

System Admin — 3/6/2026, 12:45:08 AM
Ticket created

REVIEWED

from CREATED

System Admin — 3/6/2026, 12:45:28 AM
Status changed to REVIEWED

ASSIGNED

from REVIEWED

System Admin — 3/6/2026, 4:53:28 PM
Status changed to ASSIGNED

Details

Category

ELECTRICAL

Location

PACE 15 BEEKMAN STREET

Created By

System Admin

Assigned To

Suresh Mehta

Created

3/6/2026

SLA Deadline

3/6/2026, 4:45:08 AM

Thank You